ML_Task_2(Regression)

June 5, 2026

0.1 1A) Implement Linear Regession Model on Student Marks(y) and No. of Hours(x)

[20]:

```python
# Step 1: Import libraries
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
# Step 2: Input data
x = np.array([6, 10, 2, 4, 6, 7, 0, 1, 8, 5, 3]).reshape(-1, 1)
y = np.array([82, 88, 56, 64, 77, 92, 23, 41, 80, 59, 47])

# Step 3: Create and train model
model = LinearRegression()
model.fit(x, y)

# Step 4: Get slope and intercept
m = model.coef_[0]
b = model.intercept_

# Step 5: Print equation in y = mx + b form
print(f"Equation of line: y = {m:.2f}x + {b:.2f}")

# Step 6: Predictions
Y_pred = model.predict(x)

print("\nPredicted values:")
for i in range(len(x)):
    print(f"x = {x[i][0]}, predicted y = {y_pred[i]:.2f}")

# Step 7: Predict new values
x1 = 6.5
y1 = model.predict(np.array([[x1]]))

print(f"\nPredicted y when x = {x1}: {y1[0]:.2f}")

x2 = 15
y2 = model.predict(np.array([[x2]]))
```

```
print(f"Predicted y when x = {x2}: {y2[0]:.2f}")
#x is rxc = 10x1

#y is 1 column and 10 rows

#For Multiplication , the number of columns in the first matrix must equal the␣
↪number of rows in the second matrix thats why we used reshape.
```

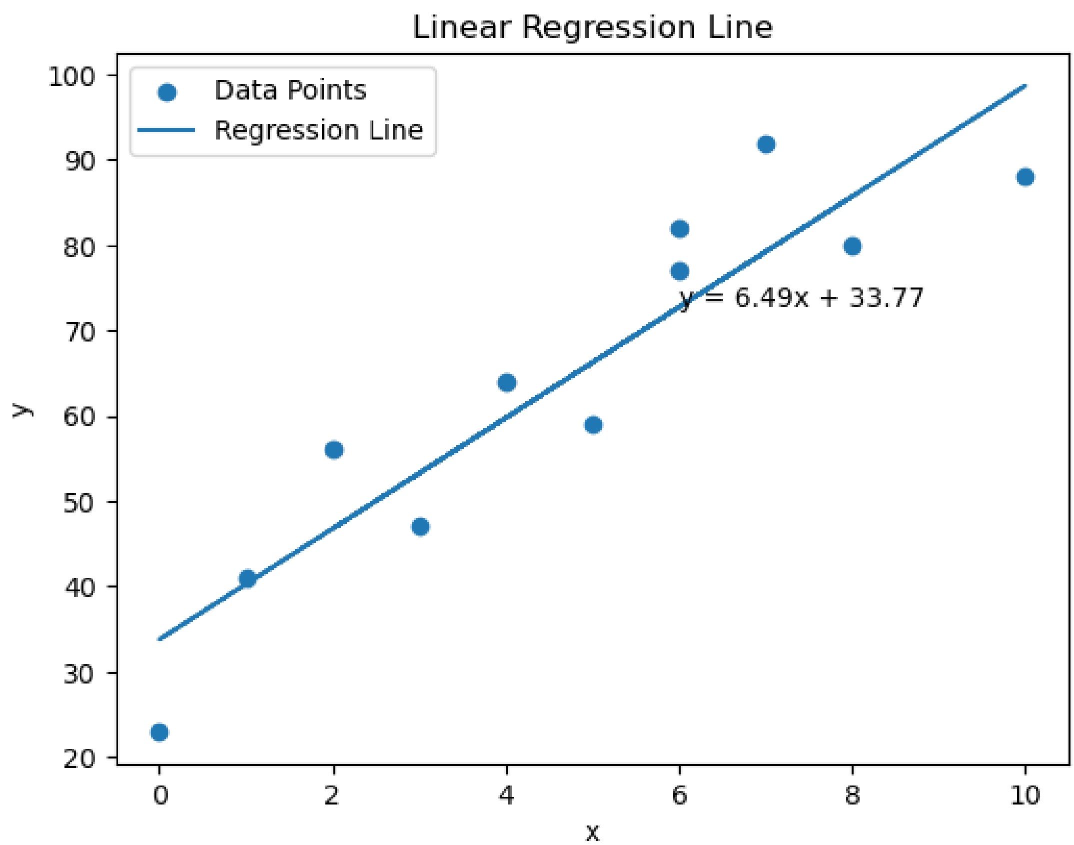
```
Equation of line: y = 6.49x + 33.77

Predicted values:
x = 6, predicted y = 72.72
x = 10, predicted y = 98.68
x = 2, predicted y = 46.75
x = 4, predicted y = 59.73
x = 6, predicted y = 72.72
x = 7, predicted y = 79.21
x = 0, predicted y = 33.77
x = 1, predicted y = 40.26
x = 8, predicted y = 85.70
x = 5, predicted y = 66.22
x = 3, predicted y = 53.24

Predicted y when x = 6.5: 75.96
Predicted y when x = 15: 131.14
```

0.2 Make Regression Line

1 Predict X on the basis of Y

[22]:
```
# Step 6: Plot scatter points
plt.scatter(x, y, label="Data Points")

# Step 7: Plot regression line
plt.plot(x, Y_pred, label="Regression Line")

# Step 8: Put equation ON the line
x_eq = 6
y_eq = m * x_eq + b

plt.text(
    x_eq,
    y_eq,
    f"y = {m:.2f}x + {b:.2f}",
    fontsize=10
)
```

```
# Step 9: Labels and title
plt.xlabel("x")
plt.ylabel("y")
plt.title("Linear Regression Line")

# Step 10: Show legend
plt.legend()
plt.savefig("linear_regression_plot.png")
# Step 11: Display graph
plt.show()
```



1.1 How many hours a student spent if graded 85 marks

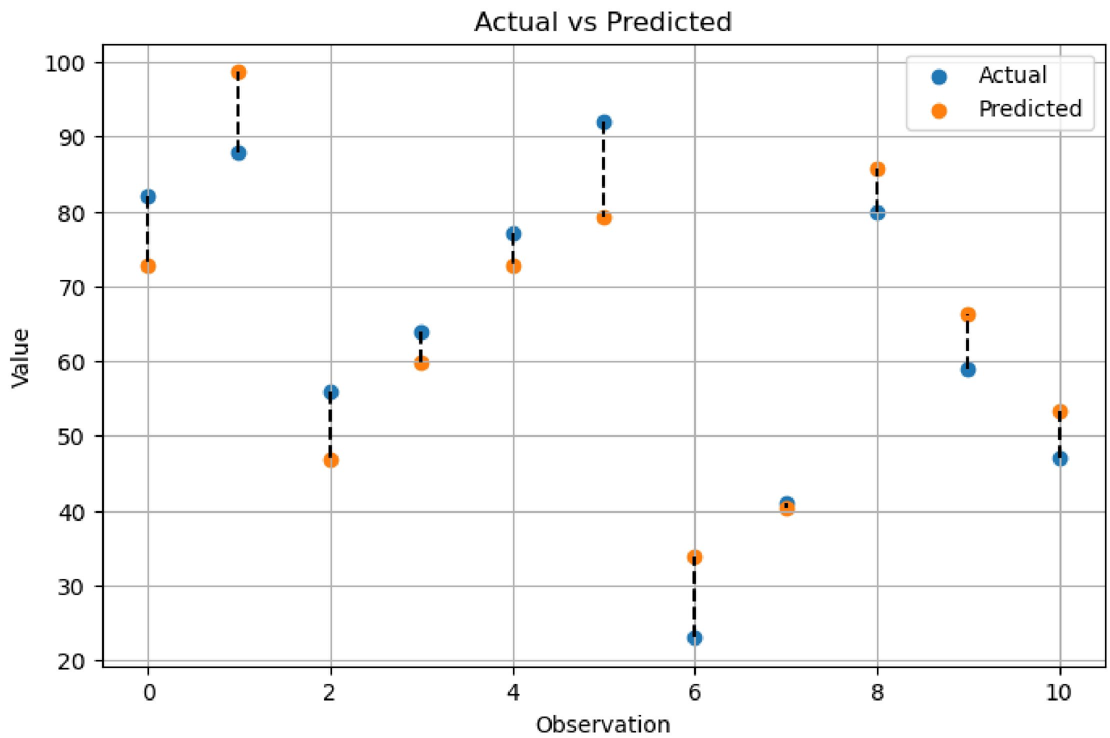
[5]:
```
# Step 1: Import libraries
import numpy as np
from sklearn.linear_model import LinearRegression

# Step 2: Input data
```

```
x = np.array([6, 10, 2, 4, 6, 7, 0, 1, 8, 5, 3]).reshape(-1, 1)
y = np.array([82, 88, 56, 64, 77, 92, 23, 41, 80, 59, 47])

# Step 3: Fit linear regression model
model = LinearRegression()
model.fit(x, y)

# Step 4: Get slope and intercept
m = model.coef_[0]
c = model.intercept_

print(f"Slope (m): {m}")
print(f"Intercept (c): {c}")

# -----------------------------
# Step 5: Find x when y = 85
# -----------------------------
y_given = 85

# Inverse prediction using same model
x_required = (y_given - c) / m

print(f"Predicted x when y = {y_given}: {x_required}")
```

```
Slope (m): 6.491312741312 7425
Intercept (c): 33.76833976833976
Predicted x when y = 85: 7.892342007434944
```

[6]:

```
import matplotlib.pyplot as plt
import numpy as np

plt.figure(figsize=(8,5))

for i in range(len(y)):
    plt.plot([i, i], [y[i], y_pred[i]], 'k--')

plt.scatter(range(len(y)), y, label='Actual')
plt.scatter(range(len(y)), y_pred, label='Predicted')

plt.xlabel("Observation")
plt.ylabel("Value")
plt.title("Actual vs Predicted")
plt.legend()
plt.grid(True)
plt.show()
```



Residual=Actual Value−Predicted Value

1.1.1 Measure R2

[7]:
```
from sklearn.metrics import r2_score

r2 = r2_score(y, y_pred)
print("R² =", r2)
```

```
R² = 0.8453212150057741
```

The coefficient of determination (R2) measures the proportion of variation in the dependent variable that is explained by the independent variable(s). An R2 value of 0.84 indicates that 84% of the variability in the response variable is explained by the regression model

2 A company has collected data on two factors(x1,x2) that may affect the sales of a product(y).

3 Perform multiple regression analysis, calculate R^2, and predict sales for: Advertising Expenditure (x1) = 75,000 Product Price (x2) = 42

[25]:
```python
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import matplotlib.pyplot as plt

# Dataset
df = pd.DataFrame({
    "Advertising_Expenditure":[50,60,45,70,80],
    "Price_of_Product":[30,35,25,40,45],
    "Sales":[500,550,450,600,650]
})

# Features and target
X = df[["Advertising_Expenditure","Price_of_Product"]]
Y = df["Sales"]

# Train model
model = LinearRegression()
model.fit(X,Y)

# Coefficients
b0 = model.intercept_
b1, b2 = model.coef_

print("Regression Model:")
print(f"y = {round(b0)} + ({round(b1)})x1 + ({round(b2)})x2")
# Predictions on training data
y_pred = model.predict(X)

# R² score
r2 = r2_score(Y, y_pred)
print(f"R² = {r2:.4f}")

# Prediction for x1=75, x2=42
new_pred = model.predict([[75000,42]])
print(new_pred[0])
```

```
Regression Model:
y = 200 + (0)x1 + (10)x2
R² = 1.0000
```

```
620.0000000001859

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739:
UserWarning: X does not have valid feature names, but LinearRegression was
fitted with feature names
  warnings.warn(
```

The X1 predictor has a coefficient very close to zero, which indicates that the prediction of y is mainly dependent on X2. The contribution of X1 to the model is negligible.

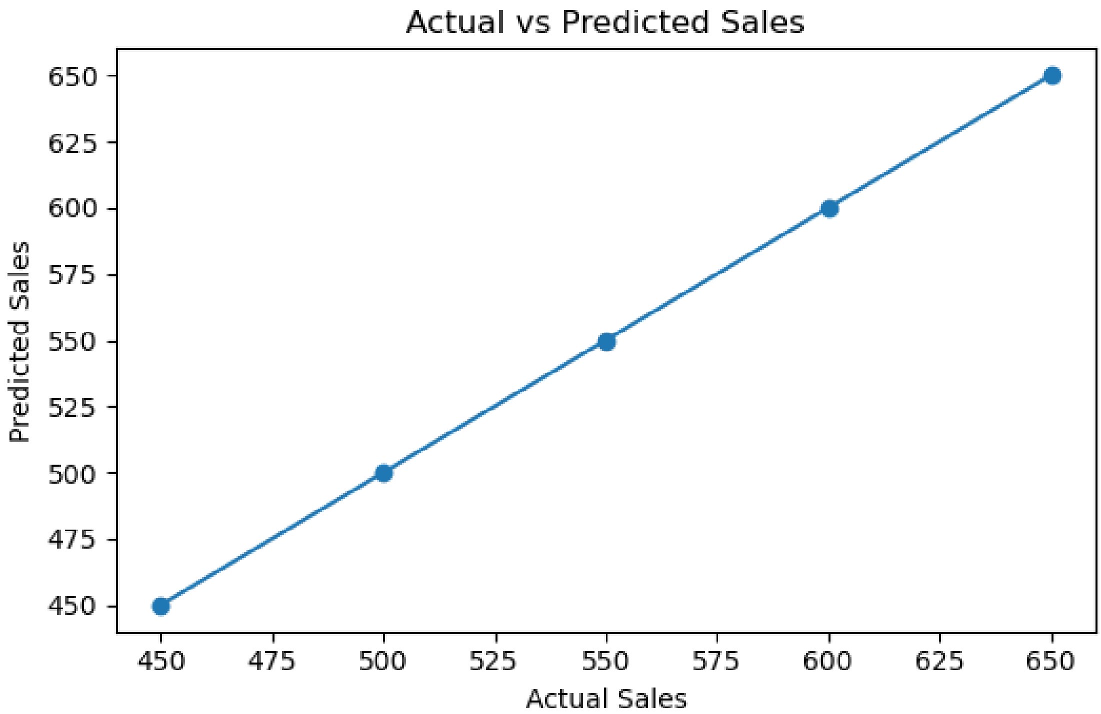
4 The estimated linear regression equation is:

4.1 ŷ = C + m1*x1* + *m2*x2

The formula to calculate slopes : ## m1= [(Σx22)(Σx1y) – (Σx1x2)(Σx2y)] / [(Σx12) (Σx22) – (Σx1x2)2]

4.2 m2 = [(Σx12)(Σx2y) – (Σx1x2)(Σx1y)] / [(Σx12) (Σx22) – (Σx1x2)2]

[27]:
```python
#Plot the Actual vs Precited Sales

import matplotlib.pyplot as plt

# Plot Actual vs Predicted
plt.figure(figsize=(6,4))
plt.scatter(Y, y_pred)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales")
plt.plot([min(Y), max(Y)], [min(Y), max(Y)])
plt.tight_layout()
#plt.savefig("sales_regression_plot.png")
```



The coefficient of determination (R2) is 1.0, indicating that 100% of the variation in Sales is explained by the predictor variables. The regression model provides a perfect fit to the data, and all predicted values match the actual values exactly.